



ADDIS ABABA
**SCIENCE AND
TECHNOLOGY**
UNIVERSITY
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Database Group Assignment

Department of Software Engineering

TITLE: HOTEL MANAGEMENT SYSTEM Using MySQL and MongoDB

Group 7

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Chapter 1: Introduction

1.1 Background of the Organization

Addis Grand Hotel is a mid-sized three-star hotel located in Addis Ababa. The hotel has approximately 120 rooms categorized into Single, Double, Deluxe, and Suite types. It provides accommodation, room service, and basic restaurant services. Currently, the hotel relies on manual record-keeping and spreadsheets to manage bookings, guest information, room status, and payments.

1.2 Problem Statement

The existing manual system at Addis Grand Hotel suffers from several limitations including frequent double bookings, difficulty in tracking real-time room availability, manual errors in billing, loss of guest records, and inefficient handling of room service orders and guest feedback. These challenges result in poor customer satisfaction, revenue leakage, and increased workload for hotel staff.

1.3 Objectives

General Objective: To develop a hybrid database system using MySQL (relational) and MongoDB (NoSQL) to automate and improve the efficiency of hotel operations.

Specific Objectives:

- To analyze the data requirements and design a normalized relational database using MySQL.
- To utilize MongoDB for storing flexible and semi-structured data such as guest reviews and room service orders.
- To enable real-time room availability checking and accurate booking management.
- To generate useful management reports (occupancy rate, revenue, etc.).
- To improve data integrity and reduce manual errors.

1.3 Scope and Limitations

The system focuses on core hotel operations: room management, guest registration, booking, payment, and basic room service. Online payment integration and mobile app development are beyond the current scope.

1.5 Benefits of the System

- Elimination of double bookings through real-time availability checking.
- Faster and more accurate check-in/check-out processes.
- Better guest feedback management using MongoDB's flexible schema.
- Improved financial reporting and revenue tracking.
- Enhanced overall customer experience.

1.6 Methodology and Tools The project followed the waterfall methodology. Tools used include:

- MySQL for structured transactional data
- MongoDB for document-oriented data
- draw.io for ER Diagram
- MySQL Workbench and MongoDB Compass for implementation
- GitHub for version control

Chapter 2: Database Design

2.1 Entities and Relationships The following main entities were identified for the Hotel Management System:

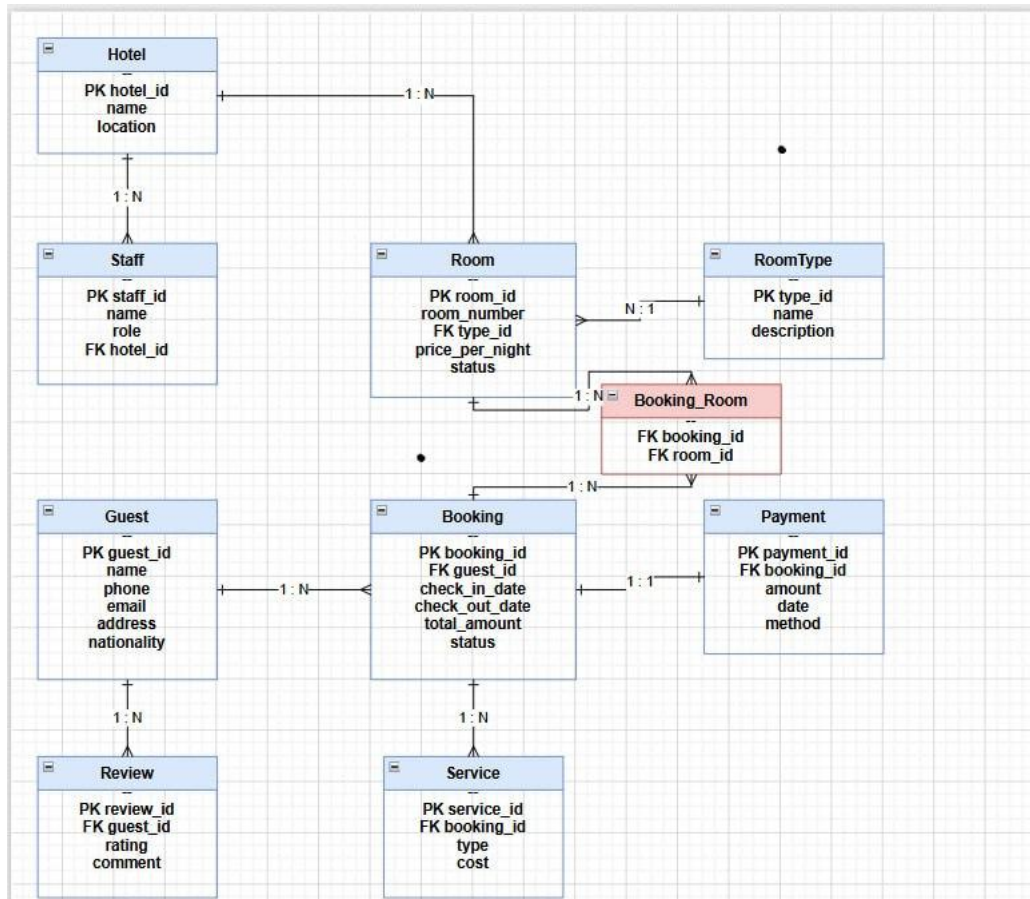
- RoomType
- Room
- Guest
- Booking
- Payment
- Staff (optional)
- Review (handled in MongoDB)
- RoomServiceOrder (handled in MongoDB)

Key Relationships:

- One Guest can make many Bookings (1:M)
- One Booking can include multiple Rooms (M:M) → resolved using BookingRoom junction table
- One Room belongs to one RoomType (M:1)
- One Booking has one or more Payments (1:M)

2.2 ER Diagram

The ER Diagram shows all entities, attributes, and relationships with proper cardinality.



2.3 Logical Schema The relational schema was designed and implemented in MySQL as follows:

- RoomType(type_id, type_name, price_per_night, max_occupancy, description)
- Room(room_id, room_number, type_id, status, floor_number)
- Guest(guest_id, full_name, phone, email, address, nationality, id_number)
- Booking(booking_id, guest_id, check_in_date, check_out_date, total_amount, status)
- BookingRoom(booking_id, room_id)
- Payment(payment_id, booking_id, amount, payment_date, payment_method)

2.4 Normalization The database was normalized up to **Boyce-Codd Normal Form (BCNF)** to eliminate redundancy and dependency issues.

Normalization Steps:

- **1NF:** Removed repeating groups and ensured atomic values.
- **2NF:** Eliminated partial dependencies.
- **3NF:** Removed transitive dependencies.
- **BCNF:** Ensured every determinant is a candidate key.

2.5 Input Forms, Output Reports Sample input forms were designed for:

- Guest Registration
- Room Booking
- Check-in/Check-out
- Room Service Order
- Payment Receipt

Key reports include:

- Room Availability Report
- Daily Revenue Report
- Guest Booking History
- Occupancy Rate Report

(Refer to Annex B for sample forms)

Chapter 3: Implementation

3.1 MySQL Implementation MySQL was used for structured and transactional data. All tables were created with proper primary keys, foreign keys, and constraints.

Schema – See mysql/schema.sql **Sample Queries** – See mysql/queries.sql

Key Features Implemented:

- Room availability checking
- Booking management with proper date validation
- Payment tracking
- Revenue and occupancy reports using JOINS and aggregations

3.2 MongoDB Implementation MongoDB was used to handle semi-structured and flexible data.

Collections:

- reviews
- roomServiceOrders
- guestFeedback

Advantages of using MongoDB:

- Flexible schema for guest reviews (ratings + comments + tags)
- Easy storage of array data (ordered items in room service)

- Fast aggregation for average ratings

Sample Documents – See mongodb/collections.json **Sample Queries** – See mongodb/queries.js

3.3 Testing and Results Both databases were tested with sample data.

- MySQL successfully handled transactional operations and complex joins.
- MongoDB efficiently managed flexible guest feedback and service orders.
- Screenshots of query results are included below.

3.4 Challenges, Bugs & Improvements

- **Challenges:** Synchronizing data between MySQL and MongoDB (guest_id and booking_id).
- **Bugs Encountered:** Date format issues in booking queries (resolved).
- **Suggested Improvements:**
 - Add user authentication and role-based access.
 - Implement real-time notifications.
 - Integrate both databases using a backend framework (Node.js / PHP).
 - Add data backup and recovery mechanisms.

References (IEEE Format)

- [1] Elmasri, R., & Navathe, S. (2016). Fundamentals of Database Systems, 7th Edition. Pearson.
- [2] MongoDB Inc. (2025). MongoDB Documentation. Available at: <https://docs.mongodb.com>
- [3] Oracle Corporation. (2025). MySQL 8.0 Reference Manual.

Annex A: Survey/Interview Questions

Project Title: Hotel Management System

Organization: Addis Grand Hotel (Mid-sized Hotel in Addis Ababa)

Section 1: Current System & Challenges

How do you currently manage room reservations and bookings?

What are the biggest problems you face with the current manual or spreadsheet system?

How do you handle guest check-in and check-out processes?

How do you track room availability and room status?

What challenges do you face with billing and payment records?

Section 2: Guest Management

6. How do you store guest information (name, contact, ID, etc.)?

7. Do guests often make special requests? How are they recorded?

8. How do you handle guest complaints or feedback?

Section 3: Room Service & Additional Services

9. How do you manage room service orders and restaurant orders?

10. How do you calculate and record payments for additional services?

Section 4: Reporting

11. What kind of reports do you need daily/weekly (occupancy, revenue, etc.)?

12. What improvements would you like to see in a new digital system?

Date: April 2026

Prepared by: [Group 7]

pull the repo in vs code and copy paste this to the file name survey_questions.pdf that is found in the folder named annex then commit and push it again

Annex B: Sample Forms

1. Guest Registration Form

Guest Full Name: _____

Phone Number: _____

Email Address: _____

Nationality: _____

ID/Passport Number: _____

Address: _____

2. Room Booking Form

Guest Name: _____

Check-in Date: _____ Check-out Date: _____

Room Type: ☐ Single ☐ Double ☐ Deluxe ☐ Suite

Number of Guests: _____

Special Requests: _____

3. Check-in / Check-out Form

Booking ID: _____

Guest Name: _____

Room Number: _____

Check-in Date & Time: _____

Check-out Date & Time: _____

4. Room Service Order Form

Room Number: _____

Guest Name: _____

Order Date: _____

Items Ordered:

Item 1: _____ Qty: ____ Price: _____

Item 2: _____ Qty: ____ Price: _____

Item 3: _____ Qty: ____ Price: _____

Total Amount: _____

Special Request: _____

5. Payment Receipt

Receipt No: _____

Guest Name: _____

Booking ID: _____

Amount Paid: _____

Payment Method: ☐ Cash ☐ Card ☐ Mobile Money

Date: _____